UME627 : Robot Dynamics And Control

L	T	P	Cr
3	0	0	3.0

Course Objective: This course will impart the knowledge on how to develop dynamic models of robot manipulators, and how to design suitable controls for robotic systems for both regulation and tracking applications. Four broad areas from robot dynamics to dynamic simulation, designing of a suitable linear as well as nonlinear controller will be discussed in detail.

Syllabus

Review of Spatial Transformation and Robot Kinematics: Spatial transformations, homogeneous transform, Euler angles, DH parameters, forward kinematics

Robot Dynamics: Acceleration of a rigid body, mass distribution and inertia tensor, Newton-Euler formulation, recursive Newton-Euler algorithm, generalized coordinates, Euler-Lagrange formulation, dynamic model in matrix form, physical significance of different forces, robot dynamics in Cartesian space, dynamic model in state-space form, dynamic modelling using bond graph

Dynamic Simulation: Dynamic simulation, forward and inverse dynamics, recursive Inverse dynamics, dynamic simulation in RoboAnalyzer/MATLAB/any equivalent software, case-studies of different manipulators

Linear Control of Manipulators: Feedback and closed-loop control, second-order systems, control of second-order systems, regulation and tracking control, control law portioning, trajectory following control, disturbance rejection

Nonlinear Control of Manipulators: Nonlinear systems**,** multivariable robot control, feedforward control, practical considerations, linearized control, individual joint PID control, Lyapunov stability analysis, computed torque control, robust control, force control

Course Learning Objectives (CLO)

The students will be able to:

1. develop the mathematical model of any robotic manipulator using Newton-Euler and Euler Lagrange approach.
2. simulate the dynamic model by developing the general code of any serial arm having any number of degrees of freedom.
3. develop a suitable linear controller for the regulation and tracking problem.
4. perform the stability analysis and develop a suitable nonlinear controller for the given task.

Text Books

1. Craig, J.J., Introduction to Robotics: Mechanics and Control, prentice Hall (2004).
2. Saha, S.K., Introduction to Robotics, McGraw Hill, Second Edition (2014).
3. Ghoshal, A., Robotics: Fundamental Concepts and Analysis, Oxford University Press (2006).
4. Pratihar, D. K., Fundamental of Robotics, Alpha Science (2016).

Reference Books

1. Fu, K.S., Gonzalez, R.C. and Lee, C.S.G., Robotics: Control, Sensing, Vision, and Intelligence, McGraw Hill (1987).
2. Schilling, R.J., Fundamentals of Robotics Analysis and Control, Prentice Hall of India (2006).
3. Niku, S.B., Introduction to Robotics: Analysis, system, application, Dorling kingsley (2006).
4. Deb, S.R. and Deb, S., Robotics Technology and Flexible Automation, McGraw Hill (2004).

Evaluation Scheme

S. No.	**Evaluation Elements**	**Weightage (%)**
1.	MST	30
2.	EST	45
3.	Sessionals (Including assignments/ Tutorials/ Quizzes etc.)	25

UME628 : Mechanics of Composite Materials

L	T	P	Cr
3	0	0	3.0

Course Objective: The objective for this course is to develop an understanding of the elastic